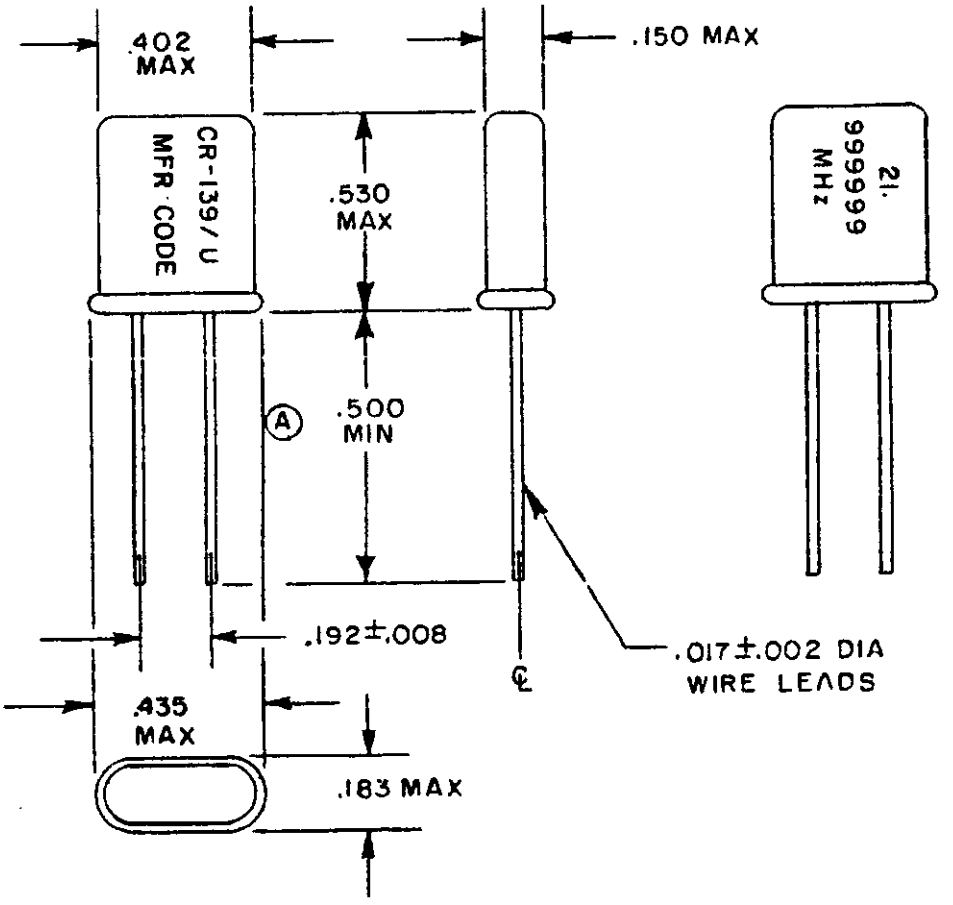
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-139/U

The complete requirements for acquiring the crystal unit, quartz, described herein shall consist of this specification sheet and the latest issue of MIL-C-3098.

Pertinent characteristics: 20 to 22 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM
.002	0.05
.008	0.20
.017	0.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.500	12.70
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Crystal unit, quartz-CR-139/U.

(A) denotes changes

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Ⓐ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking and configuration	See Figure 1
Frequency range	20 to 22 MHz, inclusive
Capacitance, shunt	12.0 pf, maximum
Mode of oscillation	Fundamental
Government reference standard test set	TS-683/TSM
Rated drive level	2.5 mw, maximum
Calibration values	Resistance 25 ohms Voltage drop 0.25 volt
Antiresonance, load capacitance	30.0 ±0.5 pf
Operating temperature range	-55° to +105°C, inclusive
Frequency tolerance	±50 ppm
Equivalent resistance	40 ohms, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Temperature cycling:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	5 ppm

1/ Method 204 of MIL-STD-202, test condition A.

NOTE: These crystal units are for use in the RTA-45A equipment. They were previously fabricated using Air Force Technical Order 12-1-20 and/or Bendix Corp specification L688450-11 thru 20.

Custodians:
Army - ER
Air Force - 85
Navy - EC

Review activities:
Air Force - 11, 99
DLA - ES

User activities:
Air Force - 17

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0623-11)

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